Niloofar Sharei

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Science Communication & Outreach

Author and Editor | Astrobites — Jan 2025 – Present
- Write and edit short-form research summaries translating peer-reviewed astrophysics papers for an audience of undergraduates and early-career researchers
- Serve on the Astrobites social media team as co-chair.

Astrobites Media Intern | AAS 248 — June 2026
- Selected as Astrobites media intern for the AAS 248 meeting (Pasadena, CA); responsibilities include organizing and publishing interviews with plenary speakers, live-blogging key conference sessions for Astrobites, staffing the Astrobites booth, and coordinating with the AAS Press Office.

Guest Author | Staryab (Farsi Astrophysical Literature Website) — Jan 2024 – Present
- Write astronomy literature summaries in Farsi.

Co-Founder & President, AURORA | UC Riverside — Jan 2024 – Present
- Founded AURORA (Astronomy Union for Reading, Outreach and Research Activities), a graduate-led organization promoting research communication and community among astronomy students.
- Organize journal club discussions, speaker events, and networking sessions; draft announcements and coordinate logistics for 15+ members.

Public Telescope Outreach | UCR Astronomy Club — Jan 2023 – Present
- Facilitate public telescope nights, explaining astronomical concepts to children and community members.
- Led gravity simulator demonstrations and interactive astronomy sessions for children ages 6–10 (Fall 2023).

Science & Engineering Fair Judge | Riverside Unified School District — Jan 2024
- Evaluated 20+ student research projects; delivered written feedback on scientific reasoning, experimental design, and communication clarity.

Education

University of California, Riverside — Sep 2022 – Present
PhD in Physics and Astronomy — *Riverside, CA*

Alzahra University
Master of Science in Theoretical Physics — *Tehran, Iran*

Sharif University of Technology
Bachelor of Science in Engineering — *Tehran, Iran*

Research Experience

Doctoral Research | University of California, Riverside
- Investigating the role of star-forming clumps in galaxy star formation rates at $0.5 < z < 3$ using *Hubble Space Telescope* imaging and SED fitting (Advisors: Prof. Brian Siana, Dr. Alexander de la Vega).

Master's Research | Alzahra University
- Stellar cluster photometry using archival HST data; constructed color-magnitude diagrams to study stellar populations (Advisor: Dr. Ahmad Shariati).

Presentations & Posters

- **Oral Presentation** – "The Dusty Truth: Do Star-Forming Clumps Survive Inward Migration?" UCR Graduate Research Symposium, Riverside, CA, April 2026. *2nd place award.*
- **Oral Presentation** – "Improving SED Fitting Assumptions in Star-Forming Clumps at $z = 0.5$–3." American Astronomical Society Meeting 247 (AAS 247), Phoenix, AZ, January 2026.
- **Poster** – "Constraints on Clump Migration from Age Gradients in CANDELS over $0.5 < z < 3$." American Astronomical Society Meeting 248 (AAS 248), Pasadena, CA, June 2026.
- **Poster** – "Stellar Clumps as Building Blocks of Galaxy Assembly." European Astronomical Society Annual Meeting (EAS 2026), Session SS5, (Remote), Switzerland, July 2026.
- **Poster** – "Role of Star-Forming Clumps in Galaxy Evolution." Santa Cruz Galaxy Workshop, UC Santa Cruz, Santa Cruz, CA, August 2025.

Publications

- **Niloofar Sharei**, A. De La Vega, B. Siana, et al., “Improving SED Fitting Assumptions in Clumps at Redshifts 0.5–3,” *in preparation.*
- A. De La Vega et al. (including **Niloofar Sharei**), “The Fraction of Clumpy Galaxies in JWST Surveys over Redshift 2–12,” *submitted to The Astrophysical Journal.* arXiv.
- W. V. Jacobson-Galán, L. Dessart, R. Margutti, et al. (including **Niloofar Sharei**), “SN 2023ixf in Messier 101: Photo-ionization of Dense, Close-in Circumstellar Material in a Nearby Type II Supernova,” *The Astrophysical Journal Letters*, 954(2), 2023. DOI.

Teaching, Mentoring & Service

Graduate Teaching Assistant | University of California, Riverside — Jan 2023 – Present

- Instruct laboratory and discussion sections for general physics courses (engineering and pre-med undergraduates, approx. 24–30 students per section); hold weekly office hours and provide individualized academic support.
- Developed reading check quizzes, pre-lab materials, and in-lab instruction; selected for the UCR Outstanding Teaching Award, Physics & Astronomy, May 2026.

Astronomy Lab Instructor | Hartnell College (via ISEE PDP) — April 2026

- Designed and co-taught an inquiry-based astronomy lab activity for community college students through the ISEE Professional Development Program (April 21–22, Salinas, CA).

Co-Instructor | UCR STEAM Academy, UC Riverside School of Education — June–July 2026

- Selected through a call open to faculty, postdocs, and graduate students to co-design and co-teach a 4-day mini-course for a cohort of 10–12 middle school students.
- Co-developed curriculum for “Escaping Heat Islands: Why Cities Get Hot, Who Gets Burned, and What to Do About It”.

GSA Career Center Liaison | UC Riverside Graduate Student Association — Feb 2026 – Present

- Elected liaison connecting graduate students to career resources, professional development programming, and UCR Career Center staff.

Physics Tutor — 2018 – 2022

- One-on-one instruction for high school students; built individualized lesson plans and adapted explanations to each student’s learning style.

Professional Development

ISEE Professional Development Program (PDP) — March – April 2026
UC Observatories / UC Santa Cruz — *Monterey & Salinas, CA*

- Completed intensive workshop training in Monterey, CA focused on inclusive, inquiry-based college STEM pedagogy.
- Applied training by designing and co-teaching an astronomy lab activity at Hartnell College, Salinas, CA (April 21–22).

Chancellor’s “Making Excellence Inclusive” Certificate Program — March – June 2026
UC Riverside Graduate Division — *Riverside, CA*

- Completing a certificate program in diversity, equity, and inclusion practices for academic and professional settings; expected completion June 2026.

Stay TA Ready Certificate Program — May 2026
UC Riverside Teaching & Learning Center (TADP) — *Riverside, CA*

- Completed professional development series on evidence-based teaching practices, inclusive pedagogy, and TA skill development.

Peer Review Skills Workshop; Proposal Writing Workshop | AAS 247 — Jan 2026
Phoenix, AZ

- Completed workshop on scientific peer review and federal research proposal writing, including communicating science to broad audiences.

AstroTech Summer School on Astronomical Instrumentation — July 2024
University of California, Berkeley — *Berkeley, CA*

- Hands-on training in optics, detectors, and astronomical instrument design.

Observing Experience

Lick Observatory | Mount Hamilton, CA May 2023 & May 2024

- 8 nights of observational training using the Shane (3m) and Nickel (1m) telescopes through the UCO Observational Astronomy Workshop.

Awards & Honors

Fellowships & Grants

- Dean's Distinguished Fellowship, University of California, Riverside.
- Tuition-waiver scholarship, M.S. in Physics, Alzahra University.
- Tuition-waiver scholarship, Sharif University of Technology.

Honors

- Outstanding Teaching Award, Physics & Astronomy Department, UC Riverside, May 2026.
- 2nd place, Oral Presentation, UCR Graduate Research Symposium, April 2026.
- Top 7% among 6,500 applicants, nationwide M.S. Physics entrance exam, Iran.
- Top 0.5% among 274,000 applicants, nationwide B.S. entrance exam (Math & Physics), Iran.
- 2nd place, province-wide High School Physics Laboratory Competition, Iran.

Travel Awards

- UCR Graduate Student Association Conference Travel Grant, $600, January 2026.
- UCR Graduate Student Association Conference Travel Grant, $500, May 2026.

Skills

Writing & Communication: Science writing (Astrobites, Staryab), Scientific presentations

Web & Digital: WordPress, Microsoft Office Suite, Canva, Adobe Creative Suite, Git/GitHub, LaTeX

Data & Programming: Python (NumPy, SciPy, Astropy, Matplotlib, Photutils), SourceExtractor, Bagpipes , Beagle, SAOImageDS9, TOPCAT, R, Matlab

Languages: English (fluent; TOEFL iBT 108/120), German (intermediate; ÖSD B1), Persian (native)